

Deep Learning for Archaeological Site Detection from LiDAR-Derived Imagery: A Multi-Version YOLO Consensus Approach

Chapter 1: Introduction and Research Objectives

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1 Introduction

The detection and documentation of archaeological sites at landscape scale remains one of the central challenges in archaeological prospection. Across Europe, and particularly in the upland regions of Scotland, thousands of archaeological features—ranging from prehistoric roundhouses and shielings to burial cairns and field systems—lie unrecorded beneath vegetation, peat, and modern land use [1]. Systematic ground survey, while accurate, is prohibitively slow and expensive when applied across entire regions. The development of airborne remote sensing technologies over the past two decades has fundamentally altered this landscape, offering the first realistic prospect of comprehensive, large-scale archaeological mapping.

Airborne Light Detection and Ranging (LiDAR) has emerged as the most transformative remote sensing technology for archaeological prospection in vegetated terrain. By emitting laser pulses that penetrate vegetation canopies and recording the reflected signal, LiDAR produces high-resolution digital terrain models (DTMs) that reveal micro-topographic features invisible to aerial photography or satellite imagery [2]. When processed through specialised visualisation techniques—including the Local Relief Model [3], Sky-View Factor [4], and multiscale hillshading—LiDAR-derived imagery can expose subtle archaeological earthworks with remarkable clarity. These visualisations have led to the discovery of tens of thousands of previously unknown sites across Europe, fundamentally reshaping our understanding of past settlement patterns and landscape use.

However, the interpretation of LiDAR-derived imagery for archaeological purposes remains overwhelmingly a manual, expert-driven process. Trained archaeologists visually inspect visualisations tile by tile, identifying and digitising features of interest. At national scales, this approach is unsustainable: Scotland alone contains approximately 80,000 km² of archaeo-

logically sensitive terrain, the majority of which has never been systematically surveyed [1]. The bottleneck is not data acquisition—LiDAR coverage is expanding rapidly through national programmes—but interpretation capacity. Manual interpretation introduces further limitations: inter-observer variability, fatigue-related omissions, and inconsistency in feature classification across different surveyors and projects.

Deep learning, and specifically convolutional neural network (CNN)-based object detection, offers a promising solution to this interpretation bottleneck. Modern object detection architectures can learn to recognise complex visual patterns directly from labelled examples, potentially matching or exceeding human performance on well-defined visual recognition tasks while operating at computational speed. In archaeology, where target features exhibit characteristic morphological signatures in LiDAR visualisations, object detection models trained on annotated examples could automate the first-pass screening of large areas, allowing human experts to focus on verification and ambiguous cases—a paradigm that has proven highly effective in medical imaging, remote sensing, and industrial inspection [5].

Among the available object detection architectures, the You Only Look Once (YOLO) family has gained particular prominence due to its combination of speed, accuracy, and accessibility. First introduced by Redmon et al. [6], YOLO formulates object detection as a single regression problem, predicting bounding boxes and class probabilities directly from image pixels in one forward pass. Successive iterations—from YOLOv5 through YOLOv8, YOLOv11, and the most recent YOLOv12—have incorporated architectural advances including anchor-free detection heads, advanced feature pyramid networks, and attention mechanisms that improve performance on small and densely clustered objects [7]. These characteristics make YOLO particularly well-suited to archaeological detection tasks, where targets are often small relative to image dimensions and occur in clusters of varying density.

This paper presents a systematic evaluation of multi-version YOLO fine-tuning for archaeological site detection in LiDAR-derived hillshade imagery, using the Isle of Arran (Scotland) as a test case. We introduce a consensus-based inference approach in which multiple random seeds of the same model architecture vote on the presence of each detected feature, requiring agreement from at least two of three seeds before a detection is retained. The study addresses three core questions: (1) How effectively do modern YOLO variants (v8, v11, v12) detect archaeological features in hillshade imagery when fine-tuned with limited labelled data? (2) Does multi-seed consensus voting improve the reliability of detections compared to single-model predictions? (3) What are the practical implications of model version choice and consensus threshold for archaeological prospection workflows?

2 Related Work

2.1 LiDAR Visualisation for Archaeological Prospection

The application of LiDAR to archaeology began in earnest in the early 2000s, with pioneering studies demonstrating its ability to penetrate forest canopy and reveal ground-surface topography at sub-metre resolution [2]. The key innovation that unlocked the archaeological potential of LiDAR was the development of specialised visualisation techniques that emphasise subtle topographic features while suppressing large-scale terrain variation.

Hesse [3] introduced the Local Relief Model (LRM), which decomposes a DTM into local (small-scale) and regional (large-scale) components through low-pass filtering, retaining only the local component to highlight archaeological features. This approach proved particularly effective for detecting low-relief earthworks such as field boundaries, trackways, and building platforms that are otherwise invisible in standard hillshade renderings. Kokalj et al. [4] proposed the Sky-View Factor (SVF), a visualisation method borrowed from atmospheric sciences that computes the proportion of visible sky from each point on a surface. SVF exaggerates convex features (banks, cairns, walls) while suppressing concave features (pits, ditches), producing visually intuitive renderings that non-specialists can interpret. Subsequent research has extended this toolkit to include multiscale hillshading, openness, positive and negative openness, and principal component analysis of multiple visualisation layers [8–10].

These visualisation techniques have been instrumental in the discovery of archaeological sites at landscape scale. In Scotland, the National Lidar Programme has acquired data covering substantial portions of the country, and projects such as the Rapid Archaeological Mapping Programme (RAMP) have demonstrated the potential of LiDAR-based survey for identifying previously unrecorded sites in upland environments [1]. Nevertheless, the interpretation process remains overwhelmingly manual, creating an urgent need for automated detection methods that can keep pace with the growing volume of available LiDAR data.

2.2 Deep Learning for Archaeological Detection

The application of deep learning to archaeological remote sensing has expanded rapidly since approximately 2018, driven by advances in computer vision, the availability of pre-trained models, and the growing volume of labelled archaeological data. Two comprehensive reviews provide a systematic overview of this emerging field.

Argyrou and Agapiou [11] reviewed 135 studies applying artificial intelligence and remote sensing to archaeology, documenting a clear shift from traditional machine learning (random forests, support vector machines) towards deep learning architectures after 2018. They identified object detection as one of the most promising application areas, with convolutional neural networks achieving superior performance on tasks ranging from burial mound detection in

satellite imagery to structure identification in drone-based orthophotography. However, they noted that the majority of studies remained at the proof-of-concept stage, applied to small geographic areas with limited ground-truth data, and that reproducibility and generalisability across different landscape contexts remained significant challenges.

Kadhim and Abed [5] provided a more technically focused review of deep learning approaches in archaeology, cataloguing the architectures, data sources, and evaluation metrics employed across 85 studies. They found that YOLO variants were among the most frequently used architectures, alongside Faster R-CNN and U-Net, with applications spanning site detection, artefact classification, and inscription recognition. Key limitations identified included the scarcity of large, publicly available annotated datasets; the computational cost of training; and the difficulty of transferring models between regions with different archaeological signatures and landscape characteristics.

2.3 Object Detection in LiDAR-Derived Imagery

Several recent studies have specifically addressed the challenge of detecting archaeological features in LiDAR-derived visualisations using deep learning, and these are most directly relevant to the present work.

Canedo et al. [12] applied a data-centric AI approach to the detection of archaeological sites in airborne LiDAR data from the Iberian Peninsula. Rather than focusing on architectural innovation, they systematically investigated the impact of data quality, annotation strategy, and training set composition on detection performance. Their results demonstrated that careful data curation—including the selection of appropriate visualisation types, consistent annotation protocols, and balanced class representation—could yield performance improvements comparable to or exceeding those from architectural changes. This finding underscored a critical insight for archaeological applications: in a domain where labelled data is scarce and expensive to produce, data-centric strategies may offer higher return on investment than model-centric ones.

Building on this work, Canedo et al. [13] developed a deep multimodal segmentation approach for the automated detection of hillforts in remote sensing imagery. Their method integrated multiple visualisation modalities—hillshade, slope, and local relief—as input channels to a segmentation network, enabling the model to learn complementary features from different representations of the same terrain. The multimodal approach outperformed single-modality baselines, particularly for features with ambiguous morphological signatures, demonstrating the value of sensor and visualisation fusion for archaeological detection.

Marçal et al. [14] provided the most direct precedent for the present study by comparing Faster R-CNN and YOLOv8 for the detection of megalithic monuments in satellite imagery. Their results showed that YOLOv8 achieved competitive accuracy with significantly faster inference speed compared to Faster R-CNN, reinforcing YOLO’s suitability for large-scale archaeological survey applications where processing speed is a practical constraint. However, their study was

limited to a single YOLO version and did not explore the impact of random seed variation or consensus-based approaches.

2.4 The Multi-Seed Consensus Gap

A notable gap in the existing literature is the absence of systematic investigation into model reliability and prediction consistency. Deep learning models are inherently stochastic: different random initialisations (seeds) can lead to different learned weights, and consequently to different predictions on the same input. In high-stakes applications such as medical diagnosis and autonomous driving, ensemble and consensus techniques are routinely employed to mitigate this variability. In archaeology, however, the standard practice remains to train a single model and report its performance, without accounting for the variability introduced by the training process.

This gap has practical consequences. An archaeological survey that relies on a single model’s predictions risks both false positives (features incorrectly identified that require unnecessary ground verification) and false negatives (genuine features missed entirely). A consensus approach, requiring agreement across independently trained models before a detection is reported, offers a principled mechanism for controlling this trade-off. To our knowledge, no prior study has systematically evaluated multi-seed consensus voting for archaeological object detection, nor compared the behaviour of different YOLO generations under this paradigm.

3 Research Objectives

The overarching aim of this study is to develop and evaluate a reproducible, multi-version deep learning pipeline for archaeological site detection in LiDAR-derived hillshade imagery, with a specific focus on model reliability through consensus-based inference. The study is conducted on the Isle of Arran, Scotland—a region with a rich archaeological record spanning the Neolithic to the post-medieval period, and for which high-resolution LiDAR-derived visualisation data are available.

The specific objectives are:

1. **To fine-tune and compare three generations of YOLO nano-scale architectures** (YOLOv8n, YOLO11n, and YOLO12n) for the detection of three archaeological feature classes—round-houses, shielings, and small cairns—in LiDAR-derived hillshade imagery. Each model version is trained under three different random seeds (0, 1, 2) to characterise the variability introduced by stochastic initialisation, producing nine independently trained models in total.
2. **To implement and evaluate a multi-seed consensus voting mechanism** in which detections from the three seeds of each model version are compared, and only those predictions for which at least two of three seeds agree (Intersection-over-Union ≥ 0.5 , same predicted class) are

retained. This approach provides a quantitative estimate of prediction reliability that goes beyond single-model confidence scores.

3. **To compare single-model, per-version consensus, and cross-version ensemble performance** using standard COCO-style mean Average Precision at 0.5 IoU threshold (mAP@0.5), supplemented by per-class precision and recall metrics, in order to characterise the trade-offs between detection sensitivity and false positive rate under different inference strategies.
4. **To establish a reproducible, open-source pipeline**—encompassing data preparation, multi-seed training, consensus inference, and quantitative evaluation—that can be applied to other archaeological contexts and LiDAR datasets with minimal adaptation [15].

By pursuing these objectives, this study contributes to the growing body of evidence on deep learning applications in archaeology while addressing the critical, and currently underexplored, issue of prediction reliability through consensus mechanisms. The results provide practical guidance for archaeological practitioners considering the adoption of automated detection methods, and for researchers developing the next generation of archaeological prospection tools.

4 Study Area and Data

The Isle of Arran (55.58°N, 5.21°W) is located in the Firth of Clyde, off the west coast of Scotland. Often described as “Scotland in miniature” due to its diverse geology and landforms, Arran contains a concentration of archaeological sites spanning from the Neolithic to the post-medieval period, including chambered cairns, standing stones, roundhouses, shielings, and extensive field systems [16]. The island has been the subject of intensive archaeological survey since the 1970s, and its archaeological record is relatively well-documented, making it an ideal test case for evaluating automated detection methods against reliable ground truth.

For this study, we use LiDAR-derived hillshade imagery of Arran processed with multidirectional illumination. The dataset comprises 424 image tiles of 500×500 pixels, manually annotated by archaeological specialists with bounding boxes for three feature classes: roundhouses (prehistoric circular stone-built dwellings, typically 8–15 m in diameter), shielings (seasonal upland pastoral structures, typically sub-rectangular, 5–10 m across), and small cairns (clearance or burial cairns, 2–6 m in diameter, often appearing as subtle mounds in the hillshade). The dataset was split into 364 training images (85.8%) and 60 validation images (14.2%), with annotations in YOLO normalised format. Instance counts per class in the training set were 260 (roundhouse), 419 (shieling), and 489 (small cairn), reflecting the relative abundance of these feature types in the Arran landscape.

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